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Analytical Cumulative Examination
September 24, 2004

1. (30%) A molecule has a Raman-active transition with a shift of 859 cm^{-1}. Determine the wavelengths in nm of the Stokes and anti-Stokes Raman lines when excited by an argon ion laser at 488.0 nm. Which is expected to be more intense and why?

2. (25%) Assume that the ratio of the anti-Stokes to Stokes Raman intensities is governed entirely by the Boltzmann distribution. Determine the ratio of intensities for the CCl_4 line at 218 cm^{-1} and 35^0C.

3. (45%) An unknown containing element X was mixed with aliquots of a standard solution of element X for atomic absorption spectroscopy. The standard solution contained 1.00 mg of X per milliliter. Data are given below:

Volume of Unknown (ML)	Volume of Standard (mL)	Total Volume (mL)	Absorbance
10.00	0.00	100.0	0.163
10.00	1.00	100.0	0.240
10.00	2.00	100.0	0.319
10.00	3.00	100.0	0.402
10.00	4.00	100.0	0.478

   a. Calculate the concentration (ug X/mL) of standard addition in each solution.
   b. Determine the concentration of X in the unknown.